JITHIN A

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CAREER SUMMARY

Software Developer/ AI Engineer with 3+ years of experience in .NET-based enterprise systems and applied AI engineering, with extensive experience in RAG, Agentic AI, and LLM-powered search and chat solutions. Passionate about continuously learning emerging AI technologies and contributing to the evolution of scalable, production-ready AI systems through thoughtful design, experimentation, and collaboration.

COURSE

1. **Certificate Programme in Generative AI – IIT Delhi (2025)**

   Completed a 6-month IIT Delhi programme covering Large Language Models (LLMs), RAG, and Prompt Engineering. Gained hands-on experience in Transformer architectures (BERT, GPT, T5), fine-tuning (LoRA, QLoRA, RLHF), and vector-database–based retrieval systems. Trained and evaluated models using PyTorch, TensorFlow, and Hugging Face, emphasizing Responsible AI, bias mitigation, and performance optimization.

GENERATIVE AI PROJECTS

1. **Fine-Tuning LLMs for Text Generation (Course Project – IIT Delhi CEP, 2025)**
   - Executed an LLM Fine-Tuning project, applying full-tuning, LoRA, QLoRA, and Prefix-Tuning to Qwen2-0.5B-Instruct on SQuAD v1.1. Performance metrics included: BLEU, ROUGE (1, 2, L), METEOR, BERT Score, Novelty, and Toxicity to ensure high-quality, safe text generation.
2. **Semantic Retrieval Using Vector Embeddings: https://github.com/Jitz99/AI-Search**
   - Built a semantic search/RAG pipeline in Python to support context-aware retrieval and generative answers.
   - Utilized vector embeddings + search index for relevance-based document matching.
3. **Context-Aware Customer Insights Chatbot: https://github.com/Jitz99/AI-Search2**
   - Developed a Structured RAG (Retrieval-Augmented Generation) system that allows users to query specific customer data using natural language.
   - Implemented a dynamic context injection pipeline: upon selecting a customer (via UI), the backend aggregates related rows from multiple relational tables (SQL) to construct a comprehensive "Customer Profile" context.
4. **Smart Health Navigator: https://github.com/Jitz99/medAssist**
   - **Medical Report Analyzer:** Developed a system that extracts clinical data from PDFs to identify normal and abnormal health results based on patient age, weight, and gender.
   - **Symptom Triage & Specialist Referral:** Built a chat interface that uses dynamic follow-up questions to assess symptoms and recommend the appropriate medical specialist by location.

SKILLS

- **Generative AI & LLMs:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Fine-Tuning (LoRA, QLoRA), Transformers (BERT, GPT, T5), NLP (spaCy, NLTK), Pytorch.
- **Agentic AI:** LangChain, LangGraph, CrewAI, OpenAI Agents SDK, Multi-Agent Systems, Tool Integration (Function Calling), Agent Workflows
- **Programming Languages**: C#, Python, TypeScript
- **Backend & APIs**: .NET 8, ASP.NET Core, RESTful APIs, Microservices
- **Databases & Vector Stores**: Azure Cosmos DB, Pinecone, SQL Server, MongoDB, ChromaDB
- **Cloud & Infrastructure:** Microsoft Azure (App Service, Functions, Azure SQL, Blob Storage,Cosmos DB), Microsoft Foundry, Docker
- **Tools & Libraries:** Hugging Face Transformers ,Scikit-learn, Pandas, Git, Azure DevOps
- **Software Engineering:** Production AI Integration, System Design, Performance Optimization, Agile Scrum

WORK EXPERIENCE

1. **Software Developer, Mariapps Marine Solutions, Kochi, India** **JAN 2025 – Present**

Key Skills: C#,.Net8 , Angular, TypeScript, Python, SQL Server, LangChain, Agent Workflows, Microsoft Azure

- **Designed and implemented a semantic search and RAG pipeline** for the QDMS Wiki using OpenAI embeddings, Azure Cosmos DB, and **LangChain**, enabling context-aware document retrieval and improving knowledge discovery over traditional keyword search.

- **Built an Agentic AI system** for vessel health inspections using **LangChain and LangGraph** to orchestrate multi-step workflows, combining RAG, SQL data retrieval, and LLM reasoning to generate vessel-specific inspection insights and recommendations.
- **Fine-tuned a Gemma 2B LLM using LoRA** to automate resume parsing and hiring automation tasks, including PII and entity extraction from unstructured resumes, **applying prompt engineering and structured output pipelines** to improve HR data processing accuracy.
- **Engineered high-performance search and application features** including Lucene-based fuzzy search and optimized Angular + C# REST APIs (infinite scroll, document sorting, HTML comparison), improving user experience and reducing page load time by **40%**.
- **Developed RESTful Web APIs** from scratch in .NET using **system design principles**, ensuring scalable, maintainable, and robust backend services.
- **Integrated Swagger (OpenAPI)** for interactive API documentation and easy testing of endpoints.
- **Leveraged GitHub Copilot** to accelerate development, improve code quality, and enhance testability.
- **Implemented new features and enhancements** following **Clean Architecture principles**, improving code modularity, testability, and maintainability.
- **Enhanced the Angular UI** by integrating custom components provided by the UI team and implementing them across project modules, improving consistency and user experience.

2. Programming Engineer, Winman Software India LLP, Mangaluru, India **SEP 2022– JAN 2025**

Key Skills: C#, Python,VB.Net,VB6,SQL Server

- **Developed and maintained enterprise features** using .NET,python and SQL Server, designing efficient database queries, stored procedures, and indexing strategies to improve data processing performance and reliability.
- **Optimized legacy application performance** using Visual Studio Profiler to identify bottlenecks and refactor data-processing logic, reducing data import time by 90% while supporting a large-scale VB6 to VB.NET migration with full functional continuity.
- **Improved code quality and maintainability** by enforcing SOLID principles, design patterns, and static code analysis, leading code reviews that reduced defect rates by 25% and strengthened engineering standards across the team.
- **Enhanced application stability** by implementing structured logging, debugging production issues, and refactoring legacy modules into cleaner, modular .NET architectures aligned with best development practices.
- **Developed a PDF parsing DLL using iTextSharp** to extract structured data from Form 16 documents, reducing manual data entry time by 70% and minimizing errors in user data updates.

EDUCATION

B.E. (Mechanical Engineering) – Sahyadri College of Engineering and Management, Mangalore 2017–2021 | CGPA: 8.37

CERTIFICATIONS

- RESTful Web API - The Complete Guide (.NET7 API) Part 1 (Udemy),2024
- Working with the OpenAI API (DataCamp),2025
- Prompt Engineering with the OpenAI API (DataCamp),2025
- Working with Hugging Face (DataCamp),2025
- LLMOps Concepts (DataCamp),2025
- Developing AI Systems with the OpenAI API (DataCamp),2025